



Quantum Cryptography and Security

Exploring quantum cryptography to secure the future from quantum threats

As we step into the quantum era, the intersection of quantum computing and cryptography unveils both unprecedented opportunities and challenges. This chapter embarks on the journey of quantum cryptography, a realm where the peculiar principles of quantum mechanics redefine the landscape of secure communication.

Quantum cryptography represents the confluence of quantum mechanics and cryptographic practices, offering security solutions that are theoretically impervious to the computational power of both classical and quantum computers. Unlike classical cryptography, which relies on the computational difficulty of certain mathematical problems, quantum cryptography is founded on the principles of quantum mechanics, ensuring